

IT ASSET RECYCLING & SCANNER AUTOMATION TOOLKIT

PowerShell automation for IT asset intake, lifecycle management, destruction tracking, and audit logging.



SCAN OR ENTER SERIAL NUMBER

Supports barcode, QR code, or manual serial number entry.



DUPLICATE DETECTION

Prevents duplicate asset records and ensures data integrity.



LIFECYCLE MANAGEMENT

Track assets through defined recycling statuses from intake to completion.



DESTRUCTION TRACKING

Record destruction status and validate destruction dates (yyyy-MM-dd).



AUDIT LOGGING

Automatic audit trail of all actions with operator accountability.



SCAN TO REGISTER

IT ASSET RECYCLING SCANNER

SELECT AN OPTION:

- [1] Scan barcode / QR code
- [2] Enter serial number manually
- [3] Exit scanner

SELECT: 1

SCAN BARCODE / QR CODE:

5CG20225PZ

ASSET REGISTERED SUCCESSFULLY

Recycling_Inventory.csv

Record ID	Asset Type	Manufacturer	Serial Number	Status	Date Scanned
REC-000001	Laptop	HP	SCG20225PZ	Ready for Recycling	2026-07-14
REC-000002	Monitor	Dell	CN0ABC12345	Recycling	2026-07-14
REC-000003	SSD	Samsung	S4EZNE0R123456	Completed	2026-07-13
REC-000004	Printer	HP	CND2K12345	Testing	2026-07-12
...	...	...	...	...	...

Scanner-Audit.csv

Timestamp	Action	Record ID	Serial Number	Status	Operator
2026-07-14 22:34	ASSET_REGISTERED	REC-000001	SCG20225PZ	Recycling	elien
2026-07-14 22:39	ASSET_RECORD_UPDATED	REC-000001	SCG20225PZ	Ready for Recycling	elien
2026-07-14 22:45	DUPLICATE_SCAN	REC-000001	SCG20225PZ	Ready for Recycling	elien
2026-07-14 23:10	ASSET_RECORD_UPDATED	REC-000001	SCG20225PZ	Completed	elien
...	...	...	...	...	...



SCAN / ENTER



REGISTER



UPDATE STATUS



AUDIT & REPORT

ELIE NZWEME

IT SUPPORT | IAM | CYBERSECURITY
POWERSHELL AUTOMATION



ABOUT THIS PROJECT

This project automates IT asset recycling operations using PowerShell and CSV to streamline asset intake, status updates, destruction tracking, and audit logging.

PORTFOLIO

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IT Asset Recycling and Scanner Automation Toolkit

Project Documentation, Workflow, Commands, and Technical Notes

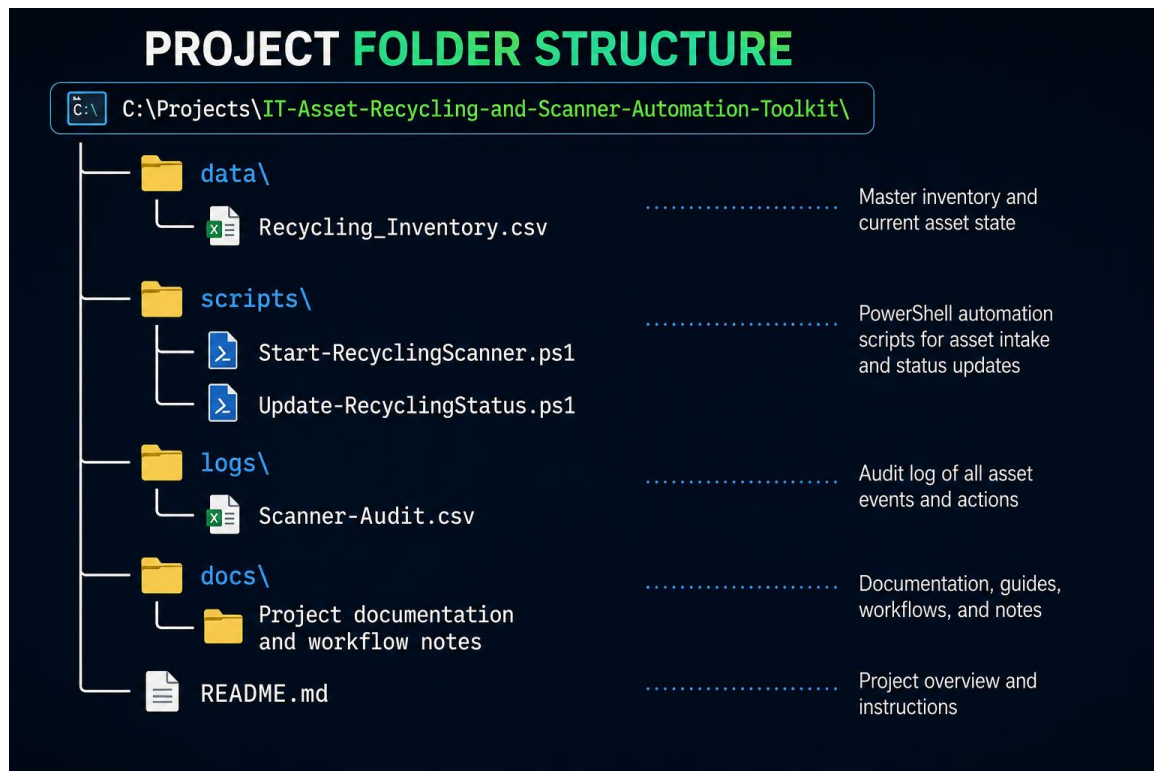
This document explains the structure and operation of the IT Asset Recycling and Scanner Automation Toolkit. The project uses PowerShell scripts, a barcode/QR scanner operating as keyboard input, and a CSV master inventory to register recycling assets, prevent duplicate serial-number records, and update recycling and destruction status.

1. Project Purpose

The purpose of this project is to automate the intake and lifecycle tracking of IT equipment being processed for recycling. Instead of manually typing complete asset records into a spreadsheet, the operator scans or enters a serial number and uses guided PowerShell menus to classify the device, select the manufacturer, and assign a recycling status.

- Register laptops, desktops, monitors, printers, storage devices, servers, phones, tablets, network devices, and other equipment.
- Capture serial numbers from barcode or QR code scans when the scanner returns the serial value.
- Allow manual serial-number entry when a device has a printed serial number without a scannable code.
- Automatically generate sequential recycling Record IDs.
- Prevent duplicate serial-number records.
- Track the current recycling lifecycle status.
- Record whether an asset was destroyed and the destruction date.
- Maintain a central CSV file that can be reviewed in Microsoft Excel.

2. Project Folder Structure



3. Folder and File Responsibilities

Location	File / Folder	Purpose
data	Recycling_Inventory.csv	Master recycling inventory and current asset state.
scripts	Start-RecyclingScanner.ps1	Registers new assets and appends new records to the inventory.
scripts	Update-RecyclingStatus.ps1	Finds an existing asset and updates status/destruction information.
logs	Scanner-Audit.csv	Stores scanner or update activity for audit and troubleshooting, when enabled by the scripts.
docs	Documentation	Stores workflow, implementation notes, screenshots, and operating procedures.
root	README.md	GitHub-facing project overview, setup, features, and usage instructions.

4. Master CSV Schema

The master inventory uses the following fields:

Field	Description
Record ID	Automatically generated unique recycling record, for example REC-000001.
Asset Type	Device classification such as Laptop, Desktop, Monitor, or Printer.
Manufacturer	Selected manufacturer or a manually entered manufacturer using Other.
Serial Number	Scanned or manually entered unique device serial number.
Status	Current position in the recycling workflow.
Date Scanned	Date the asset was registered, stored in yyyy-MM-dd format.
Was Asset Destroyed	Yes or No destruction indicator.
Date Destroyed	Manually entered destruction date in yyyy-MM-dd format; blank until applicable.

5. Recycling Status Values

- Received – Asset has entered the recycling intake process.
- Testing – Asset is being inspected or tested.
- Data Wipe Pending – Data-bearing equipment requires sanitization.
- Ready for Recycling – Processing is complete and the asset is ready for recycling.
- Recycling – Asset is currently in the recycling process.
- Refurbish – Asset may be repaired or reused.
- Parts – Asset is being retained or dismantled for parts.
- Completed – Recycling/destruction workflow is complete.

6. PowerShell Preparation and Commands

The project was run from the following project directory:

```
cd C:\Projects\IT-Asset-Recycling-and-Scanner-Automation-Toolkit
```

When PowerShell blocked local script execution, the following command was used to bypass the execution policy for the current PowerShell process only:

```
Set-ExecutionPolicy -Scope Process Bypass
```

Scope Process means the change applies only to the current PowerShell session. Closing that PowerShell window ends the temporary bypass. This is preferable for the lab/project workflow because it does not permanently change the execution policy for the entire computer.

The scanner registration script is started with:


```
.\scripts\Start-RecyclingScanner.ps1
```

The existing-record status update script is started with:

```
.\scripts\Update-RecyclingStatus.ps1
```

7. New Asset Registration Workflow

The Start-RecyclingScanner.ps1 workflow operates as follows:

1. The script starts and identifies Recycling_Inventory.csv as the master inventory.
2. The cursor waits at SCAN / SERIAL NUMBER.
3. The operator scans a barcode or QR code that returns a serial number, or manually types the serial number.
4. The script normalizes and checks the serial number against existing inventory records.
5. If the serial number already exists, the device is identified as a duplicate and is not added again.
6. If the serial number is new, the operator selects the Asset Type.
7. The operator selects a Manufacturer from the suggested list. If the manufacturer is not listed, Other allows manual entry.
8. The operator selects the asset Status.
9. The script determines the next sequential Record ID, such as REC-000001 or REC-000002.
10. The script automatically sets Date Scanned using yyyy-MM-dd.
11. Was Asset Destroyed is initially stored as No and Date Destroyed remains blank for a newly registered asset.
12. The new object is appended to Recycling_Inventory.csv.
13. The script displays ASSET REGISTERED and returns to SCAN NEXT DEVICE for continuous intake.

8. Scanner Input Behavior

The handheld scanner acts as an input device. When a barcode or QR code contains the device serial number, the scanner types that value into the active PowerShell prompt and normally submits it using its configured Enter/Return suffix. The PowerShell script does not magically discover manufacturer or asset type from an arbitrary serial number. Those fields are selected through guided menus to keep the recycling record accurate.

For equipment with a serial number printed as plain text and no supported scannable code, the operator can type the serial number manually at the same SCAN / SERIAL NUMBER prompt.

9. Duplicate Prevention

Before a new record is created, Start-RecyclingScanner.ps1 checks the Serial Number field in Recycling_Inventory.csv. If a matching serial number is found, the script displays a duplicate device warning and does not append another row.

```
DUPLICATE DEVICE DETECTED  
Serial Number: 5CG20225PZ  
Record ID: REC-000001  
Device: HP Laptop  
Status: Recycling
```

```
DEVICE NOT ADDED
```

Serial Number is therefore treated as the primary duplicate-control value, while Record ID is the toolkit's internally generated tracking identifier.

10. Existing Asset Update Workflow

Update-RecyclingStatus.ps1 is used after an asset already exists in the CSV.

1. Run the update script.
2. Scan or enter the Serial Number or Record ID.
3. The script imports Recycling_Inventory.csv and locates the matching record.
4. The current Record ID, device, serial number, status, destruction flag, and destruction date are displayed.
5. Select the new recycling Status.
6. Answer whether the asset was destroyed using Y or N.
7. If Y is selected, manually enter Date Destroyed in yyyy-MM-dd format.
8. The script validates the date format.
9. The matching in-memory CSV record is modified.
10. The inventory is exported back to the same Recycling_Inventory.csv file.
11. The script displays RECORD UPDATED with the new values.

11. Date Destroyed Validation

Date Destroyed is manually entered. The required structure is:

yyyy-MM-dd

Example:

2026-07-14

The date validation prevents inconsistent formats such as 7/14/26 or July 14, 2026 from being stored when the script expects the standardized project format.

12. Example End-to-End Record

Initial registration:

Record ID:	REC-000001
Asset Type:	Laptop
Manufacturer:	HP
Serial Number:	5CG20225PZ
Status:	Recycling
Date Scanned:	2026-07-14
Was Asset Destroyed:	No
Date Destroyed:	

After running Update-RecyclingStatus.ps1:

Record ID:	REC-000001
Status:	Completed
Was Asset Destroyed:	Yes
Date Destroyed:	2026-07-14

13. CSV Update Behavior

Start-RecyclingScanner.ps1 adds new rows to the master inventory. Update-RecyclingStatus.ps1 does not create a replacement asset record when updating a device. It finds the existing record and changes the applicable fields before saving the inventory back to the same CSV file.

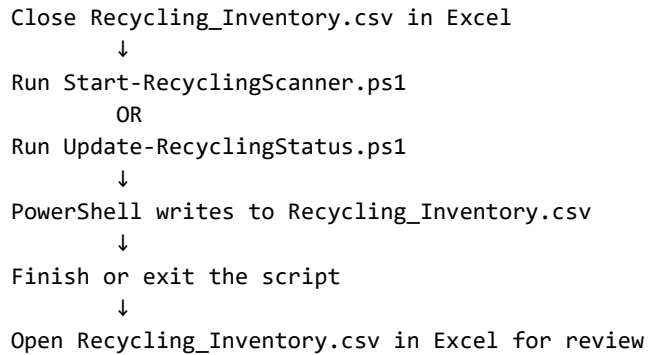
The same Recycling_Inventory.csv file is therefore the project's source of truth for both new asset registration and lifecycle status updates.

14. CSV File Lock Issue and Resolution

A save error occurred when Recycling_Inventory.csv was open in Microsoft Excel. Excel can hold a lock on the CSV file, preventing PowerShell Export-Csv from replacing or rewriting it.

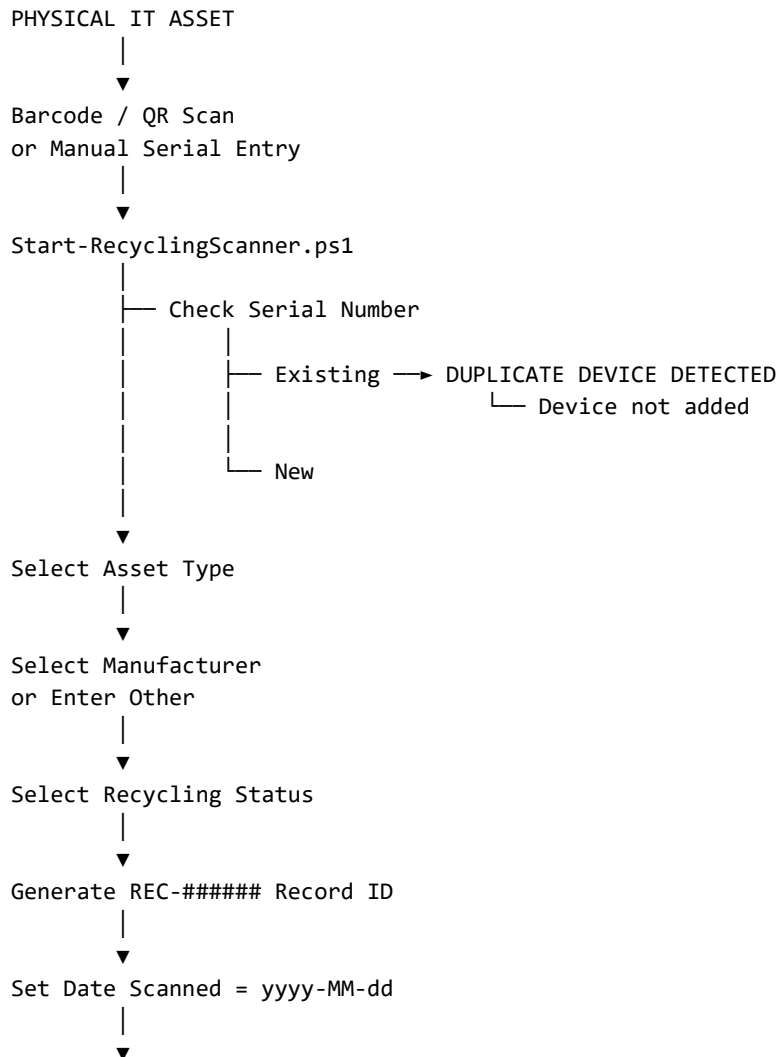
The process cannot access the file '...\Recycling_Inventory.csv' because it is being used by another process.

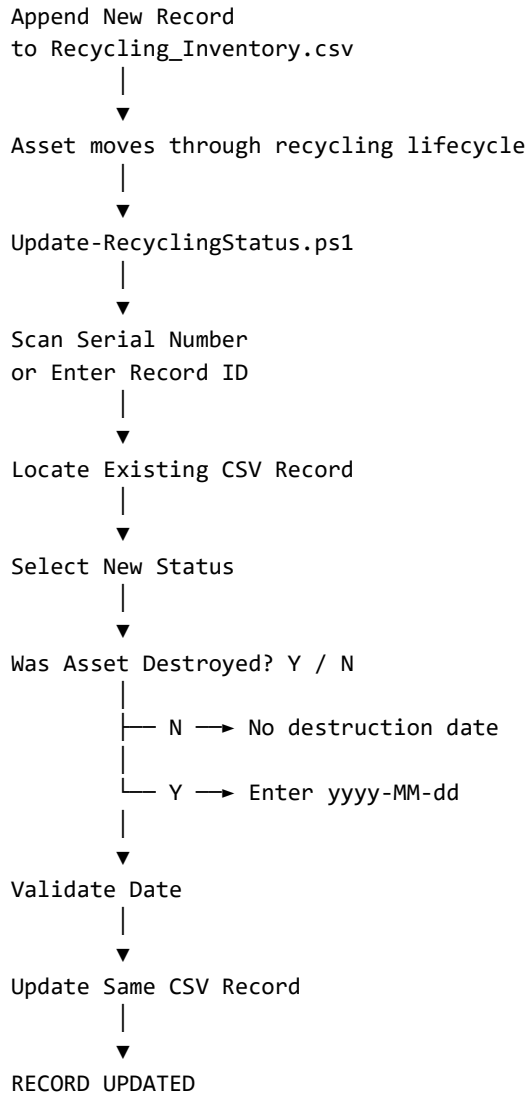
Recommended operating workflow:



The updated status script includes file-lock handling so the operator can be warned when the CSV is in use. The inventory should still be closed in Excel before running write operations.

15. Complete Technical Workflow





16. Daily Operating Procedure

For new recycling intake:

```
cd C:\Projects\IT-Asset-Recycling-and-Scanner-Automation-Toolkit
Set-ExecutionPolicy -Scope Process Bypass
.\scripts\Start-RecyclingScanner.ps1
```

For lifecycle or destruction updates:

```
cd C:\Projects\IT-Asset-Recycling-and-Scanner-Automation-Toolkit
Set-ExecutionPolicy -Scope Process Bypass
.\scripts\Update-RecyclingStatus.ps1
```

To exit the scanner workflow, type:

EXIT

17. Project Skills Demonstrated

- PowerShell scripting and interactive menu design.
- IT asset lifecycle and recycling workflow automation.
- Barcode and QR scanner integration through keyboard-emulation input.
- CSV data ingestion, object manipulation, and export.
- Duplicate-record prevention and data-quality controls.
- Sequential identifier generation.
- Input and date validation.
- File-lock awareness and operational error handling.
- Lifecycle status tracking and destruction documentation.
- Process documentation and repeatable operating procedures.

18. Recommended GitHub Documentation Evidence

- Project architecture and folder structure.
- README with setup and execution commands.
- Screenshots of the scanner intake workflow.
- Screenshot of duplicate-device detection.
- Screenshot of a successful asset registration.
- Screenshot of Update-RecyclingStatus.ps1 finding an existing record.
- Screenshot of a Completed and destroyed asset update.
- A sanitized sample Recycling_Inventory.csv with fake serial numbers.
- A workflow diagram and explanation of the source-of-truth CSV.
- A note explaining that real company asset data and serial numbers should not be published publicly.

19. Summary

The IT Asset Recycling and Scanner Automation Toolkit provides a lightweight, repeatable asset recycling workflow. New devices are registered through Start-RecyclingScanner.ps1, existing records are maintained through Update-RecyclingStatus.ps1, and Recycling_Inventory.csv acts as the central source of truth. The project demonstrates practical PowerShell automation, asset lifecycle tracking, data validation, duplicate prevention, and operational documentation.